# Project Brief: Fernhill Software, version 1 of 3 (the enthusiast's model)

## 1. The organization

Fernhill Software is a useful fiction, a business-software firm with about 1,100 engineers in 120 product teams. It hires engineers on technical skill and promotes its strongest into management, where some struggle. In 2025, as AI coding tools arrive, an AI enthusiast on the leadership team proposes a three-part reorganization. First, hire professional managers on people skill instead of promoting engineers. Second, make each team's engineer-manager its technical lead, keeping technical judgment but managing no one. Third, let AI fill the manager's technical gaps. **Fernhill adopts the plan** in all 120 teams in July 2025, with AI at the same time.

## 2. The model

A team delivers when two jobs get done, coordinating people and resolving technical unknowns. The proposal splits those jobs between a manager and a lead, with AI assisting both.

1. **People skill lowers coordination friction.** Less work sits blocked on other people, more so in bigger teams.
2. **AI covers the manager's technical gaps.** It explains technical status, checks estimates, and drafts plans.
3. **Technical judgment stays with the lead**, who resolves the team's unknowns as the engineer-manager did, so nothing technical is lost.

**In short,** the reorganization puts people skill in the manager's chair without losing technical judgment, so delivery rises.

**For the model to be right,** the professional managers must have more people skill than the engineer-managers they replace. AI must cover the manager's technical gap on hard work as well as easy. The leads must accept their new role. Nothing else that affects delivery may change at the same time. **What stays unmeasured** is whether AI actually covers the manager's gap, since licenses show only access.

## 3. The question

Should Fernhill replace engineer-managers with professional managers backed by technical leads and AI? The rollout answers with one comparison. It compares **before and after** July 2025 across all 120 teams, which shows the whole package as adopted, if nothing else changed.

## 4. The prediction

Stated before looking at this version's data:

- **Delivery** rises after July 2025, by about 5 percentage points, after a dip in the first quarter.
- **People-skill ratings** rise after the reorganization.
- The rise **holds on hard quarters** as well as easy ones, because AI covers the technical gap.

The model is wrong if any of these holds:

- Delivery is no higher after the reorganization than before.
- People-skill ratings rise but delivery doesn't. Then people skill isn't what drives delivery.
- Delivery rises on easy quarters but falls on hard ones. Then AI doesn't cover the technical gap.

## 5. The data sketch

**The data.** Fernhill's records have one row per team per quarter, July 2023 to June 2026 (1,440 rows), and they are simulated, since Fernhill is fictional. **The variables,** by concept. The change is before or after July 2025, manager type, and AI licenses. The manager has a people-skill rating on the eight Project Oxygen behaviors, and the hours work sits blocked on other teams. The work has difficulty, rated at the quarter's close. The outcome is the share of committed milestones delivered on time.

## 6. The disciplinary anchor

**Whom to promote (organizational management).** Firms promote their best performers, who then make worse managers (Peter and Hull, 1969; Benson, Li and Shue, 2019). People skill keeps teams together, and Google's Project Oxygen ranked technical skill last of eight manager behaviors (Hoffman and Tadelis, 2021; Garvin, 2013).

**The work itself (operations management).** Cheaper knowledge means fewer problems go up the hierarchy, and AI helps less experienced workers most (Garicano, 2000; Brynjolfsson, Li and Raymond, 2025).

**The method.** The comparison is before and after for the whole firm, with eight quarters of history as the baseline.

## Note on AI use

I used Gemini (Willamette account) to turn the leadership proposal into relationships and match each concept to a record. It suggested checking for other 2025 changes that could move delivery. I listed that as an assumption, since the customer migration is a workload change, not a management change. It suggested rating difficulty at planning. I rejected that, since the quarter-close review already rates every quarter.